

# Introduction to Internet of Things (IoT)

## Learning Objectives

### Introduction to IoT:

- Architectural overview,
- Design principles and needed capabilities,
- IoT Applications,
- Sensing, Actuation,
- Basics of Networking, M2M and IoT Technology
- Fundamentals-Devices and gateways, Data

management,

□ Business processes in IoT, Role of cloud in IoT,  
Define IoT

## What is IoT?

- IoT stands for Internet of Things. It refers to the interconnectedness of physical devices, such as – appliances and vehicles, that are embedded with software, sensors, and connectivity
  - which enables these objects to connect and exchange data.
- Main components used in IoT:
  - Low-power embedded systems: Less battery consumption, high performance are the inverse factors that play a significant role during the design of electronic systems.

- Sensors: Sensors are the major part of any IoT application. It is a physical device that measures and detects certain physical quantities and converts it into signal which can be provided as an input to processing or control unit for analysis purpose.

## IoT Sensors

Different types of Sensors:

1. Temperature Sensors
2. Image Sensors
3. Gyro Sensors
4. Obstacle Sensors
5. RF Sensor
6. IR Sensor
7. MQ-02/05 Gas Sensor

8. LDR Sensor

9. Ultrasonic Distance Sensor

# IoT Applications

## **Modern Applications:**

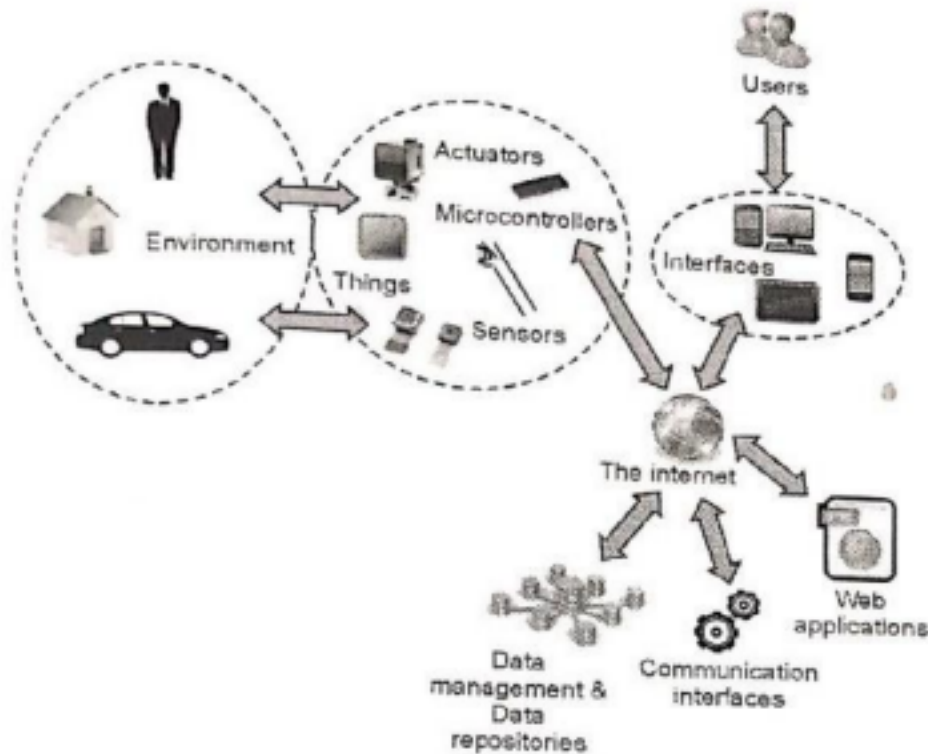
1. Smart Grids and energy saving
2. Smart cities
3. Smart homes/Home automation
4. Healthcare
5. Earthquake detection
6. Radiation detection/hazardous gas detection
7. Smartphone detection
8. Water flow monitoring
9. Traffic monitoring
10. Wearables
11. Smart door lock protection system
12. Robots and Drones
13. Healthcare and Hospitals, Telemedicine applications

14. Security

15. Biochip Transponders (For animals in farms)

# Evolution of IoT

- Internet → Mobile Internet → Internet of Things



# Main Components of IoT

- Sensors

- Control Units
- Cloud Computing
- Networking
- Embedded Systems

## Advantages

### **Advantages of IoT:**

1. Improved efficiency and automation of tasks.
2. Increased convenience and accessibility of information.
3. Better monitoring and control of devices and

systems.

- 4. Greater ability to gather and analyze data.
- 5. Improved decision-making.
- 6. Cost savings.

## Disadvantages

### **Disadvantages of IoT:**

- 1. Security concerns and potential for hacking or data breaches.
- 2. Privacy issues related to the collection and use of personal data.
- 3. Dependence on technology and potential for system failures.
- 4. Limited standardization and interoperability among devices.
- 5. Complexity and increased maintenance requirements.
- 6. High initial investment costs.
- 7. Limited battery life on some devices.
- 8. Concerns about job displacement due to automation.
- 9. Limited



regulation and legal framework for IoT, which can lead to confusion and uncertainty.

# IoT Architecture

- Internet of Things (IoT) technology has a wide variety of applications and use of Internet of Things is growing so faster.
- Depending upon different application areas of Internet of Things, it works accordingly as per it has been designed/developed.
- But it has not a standard defined architecture of working which is strictly followed universally. • So. here in this we will discuss basic fundamental

architecture of IoT i.e., 4 Stage IoT architecture.

## IoT Architecture

